(X) We don't use b−a for arc length because it only measures straight line (horizontal) distance. Arc length formulas consider the curve's shape, adding up tiny segment along the curve to get the true path length.

Topic 6-3

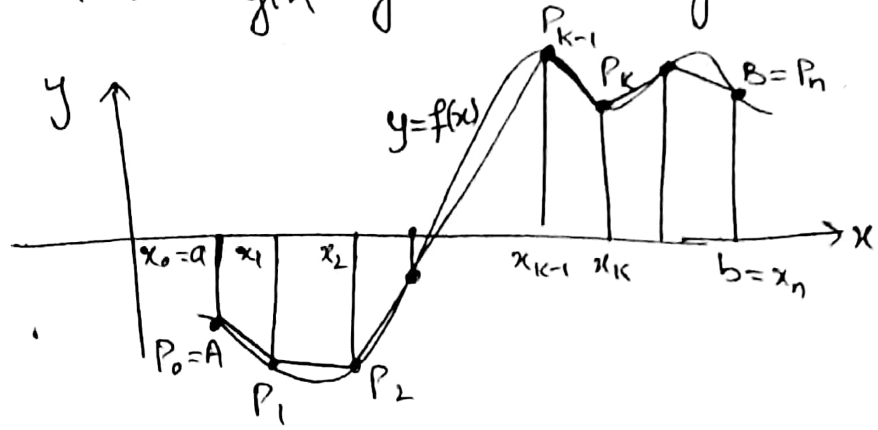
Arc Length

Arc length is the distance along a curve line between two points.
• It differs from simple distance because it considers the curve's shape.

Objective: To understand the concept of arc length of the curve, derive its formula and see its differential form.

Length of the curve: To approximate the length of a Smooth* curve $y = f(x)$ we take a partition and sum the length of the line segments.

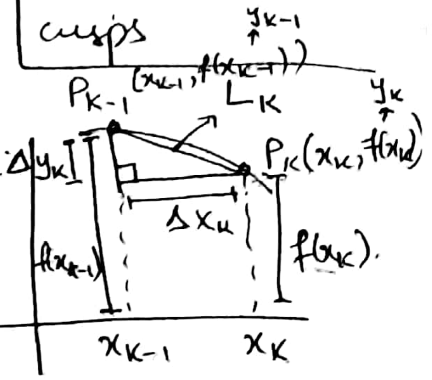
*If a function f has a cont. derivative at every point of $[a,b]$. Such a function is called smooth and its graph is a smooth curve because it does not have any breaks, corners or cusps.



Riemann Sum Approximation:

The length of a segment is:

$$P_{k-1}P_k = \sqrt{\Delta x_k^2 + \Delta y_k^2}$$



It is close to the tangent line

So $\Delta y_k \approx f'(x_k)\,\Delta x_k$ (∵ Mean Value Theorem (Topic 4.2))

$$\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \approx \sqrt{(\Delta x_k)^2 + (f'(x_k)\,\Delta x_k)^2}$$

$$= \sqrt{(\Delta x_k)^2\left[1 + (f'(x_k))^2\right]}$$

$$= \sqrt{1 + (f'(x_k))^2}\,\Delta x_k \rightarrow$$ This is the length formula for one segment.

$$\lim_{n\to\infty} \sum_{k=1}^{n} \sqrt{1 + (f'(x_k))^2}\,\Delta x_k = \boxed{\int_a^b \sqrt{1 + [f'(x)]^2}\,dx}$$ → This is length formula for whole curve.

Definition: If f' is cont. on $[a,b]$ then the length (arc length) of the curve $y=f(x)$ from point $A(a, f(a))$ to the point $B(b, f(b))$ is the value of the integral,

$$\boxed{L = \int_a^b \sqrt{1+[f'(x)]^2}\,dx = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx.}$$

If f' is not cont. on $[a,b]$ then we'll use another formula to find arc length.

Formula for the Length of $x=g(y)$, $c \leq y \leq d$:

If g' is cont. on $[c,d]$, the length of the curve $x=g(y)$ from $A=(g(c), c)$ to $B=(g(d), d)$ is

$$\boxed{L = \int_c^d \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy = \int_c^d \sqrt{1+[g'(y)]^2}\,dy}$$

Example 1 (From book) – DIY

Example 2 Find the length of the graph of

$$f(x) = \frac{x^3}{12} + \frac{1}{x} \qquad 1 \leq x \leq 4$$

Sol:

$$f'(x) = \frac{3x^2}{12} + \left(-\frac{1}{x^2}\right) = \frac{x^2}{4} - \frac{1}{x^2}.$$

$$1+[f'(x)]^2 = 1+\left(\frac{x^2}{4} - \frac{1}{x^2}\right)^2 = 1 + \frac{x^4}{16} - \frac{1}{2} + \frac{1}{x^4}$$

$$= \frac{x^4}{16} + \frac{1}{x^4} + \frac{1}{2} = \left(\frac{x^2}{4} + \frac{1}{x^2}\right)^2$$

$$L = \int_1^4 \sqrt{\left(\frac{x^2}{4}+\frac{1}{x^2}\right)^{\cancel{2}}}\,dx \qquad \left(\because L = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx\right)$$

$$L = \int_1^4 \left(\frac{x^2}{4} + \frac{1}{x^2}\right) dx$$

$$L = \frac{x^3}{12}\Big|_1^4 - \frac{1}{x}\Big|_1^4 = \left(\frac{64}{12} - \frac{1}{12}\right) - \left(\frac{1}{4} - 1\right)$$

$$= \frac{72}{12} = \boxed{6} \quad \text{Ans}$$

<u>Example 3</u> Find the length of curve $y = \left(\frac{x}{2}\right)^{2/3}$ from $x = 0$ to $x = 2$.

<u>Sol</u>:

$$y' = \frac{\cancel{2}}{3}\left(\frac{x}{2}\right)^{2/3 - 1}\left(\frac{1}{\cancel{2}}\right) = \frac{21}{3}\left(\frac{x}{2}\right)^{-1/3}\left(\frac{1}{\ }\right.$$

$$y' = \frac{1}{3}\left(\frac{2}{x}\right)^{1/3}$$

y' is not defined at $x = 0$ So we cannot find the curve's length with $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

We therefore rewrite the eq. to express x in terms of y

$$y = \left(\frac{x}{2}\right)^{2/3}$$

$$y^{3/2} = \frac{x}{2}$$

$$x = 2y^{3/2}$$

When $x = 0 \implies y = 0$.

When $x = 2 \implies y = 1$.

$$x' = \cancel{2}\left(\frac{3}{\cancel{2}} y^{3/2 - 1}\right) = 3y^{1/2}.$$

$$\sqrt{1 + (x')^2} = \sqrt{1 + (3y^{1/2})^2} = \sqrt{1 + 9y}.$$

$$L = \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dx = \int_0^1 \sqrt{1 + 9y}\, dy$$

$$= \frac{1}{9}\int_0^1 \sqrt{1 + 9y}\cdot 9\,dy$$

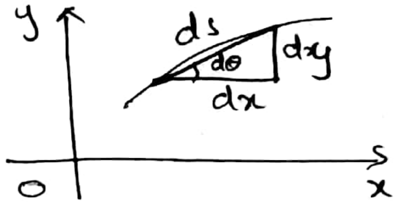
$$= \frac{1}{9}\frac{(1 + 9y)^{1/2 + 1}}{1/2 + 1}\Big|_0^1 = \frac{2}{27}(1 + 9y)^{3/2}\Big|_0^1$$

$$= \frac{2}{27}\left(10\sqrt{10} - 1\right) \approx \boxed{2.27}$$

Ans

Trick to remember: The total length is the integral of the infinitesimal (infinitely many) displacement ds along the curve.

$$ds^2 = dx^2 + dy^2$$
$$ds = \sqrt{dx^2 + dy^2}$$



Distance is the total length of the path b/t two points

Displacement is the shortest straight line distance from starting to endpoint.

	$y = f(x)$	$x = g(y)$
ds	$\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$	$\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$

The Differential formula for Arc length:

If $y = f(x)$ and if f' is cont. on $[a,b]$ then by Fundamental Theorem of Calculus (Part 1) we can define a new function

$$S(x) = \int_a^x \sqrt{1+[f'(t)]^2}\,dt.$$

The function S is called the arc length function for $y = f(x)$. From the Fundamental Theorem, the function S is differentiable on (a,b) and

$$\frac{ds}{dx} = \sqrt{1+[f'(x)]^2} = \sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

Then the differential of arc length is

$$ds = \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx.$$

Example 4: Find the arc length function for the curve in Example 2 taking $A = \left(1, \frac{13}{12}\right)$ as the starting point.

Sol: $1+[f'(x)]^2 = \left(\frac{x^2}{4}+\frac{1}{x^2}\right)^2$

$$S(x) = \int_1^x \sqrt{1+[f'(t)]^2}\, dt = \int_1^x \sqrt{\left(\frac{t^2}{4}+\frac{1}{t^2}\right)^2}\, dt$$

$$= \int_1^x \left(\frac{t^2}{4}+\frac{1}{t^2}\right) dt = \frac{t^3}{12}\Big|_1^x + \left(-\frac{1}{t}\right)\Big|_1^x$$

$$= \frac{x^3}{12} - \frac{1}{12} - \frac{1}{x} + 1$$

$$= \boxed{\frac{x^3}{12} - \frac{1}{x} + \frac{11}{12}}. \text{ Ans}$$

$\longleftrightarrow$

Ex 6.3

Find the length of the curves.

Q1 $y = \frac{1}{3}(x^2+2)^{3/2}$ $x=0$ to $x=3$.

Sol: $y' = \frac{1}{\cancel{3}} \cdot \frac{\cancel{3}}{\cancel{2}}(x^2+2)^{1/2}(\cancel{2}x)$

$(y')^2 = x^2(x^2+2) = x^4+2x^2$

$1+(y')^2 = 1+x^4+2x^2 = (x^2+1)^2$.

$$L = \int_0^3 \sqrt{(x^2+1)^{\cancel{2}}}\, dx = \int_0^3 (x^2+1)\, dx = \frac{x^3}{3}\Big|_0^3 + x\Big|_0^3$$

$$= \frac{3^3}{3} - 0 + (3-0)$$

$$= \boxed{12} \text{ Ans}$$

$\longleftrightarrow$

Q2 $y = x^{3/2}$ $\quad 0 \le x \le 4.$

Sol: $y' = \frac{3}{2} x^{1/2}$

$(y')^2 = \left(\frac{3}{2} x^{1/2}\right)^2 = \frac{9}{4} x.$

$1 + (y')^2 = 1 + \frac{9}{4} x.$

$$L = \int_0^4 \sqrt{1 + \frac{9}{4}x}\, dx = \frac{4}{9}\int_0^4 \left(1 + \frac{9}{4}x\right) \frac{9}{4}\, dx$$

$$= \frac{4}{9} \frac{\left(1 + \frac{9}{4}x\right)^{3/2}}{3/2} \Bigg|_0^4 = \frac{8}{27}\left[\left(1 + \frac{9}{4}(4)\right)^{3/2} - \left(1 + \frac{9}{4}(0)\right)^{3/2}\right]$$

$$= \boxed{\frac{8}{27}\left(10^{3/2} - 1\right)} \quad \text{Ans}$$

Q3 $x = \frac{y^3}{3} + \frac{1}{4y}$ $\quad y = 1$ to $y = 3.$

Sol: $L = \int_1^3 \sqrt{1 + (x')^2}\, dy$

$$x' = \frac{\cancel{3} y^2}{\cancel{3}} - \frac{1}{4y^2} = y^2 - \frac{1}{4y^2} = \frac{4y^4 - 1}{4y^2}$$

$$(x')^2 = \frac{16y^8 + 1 - 8y^4}{16y^4}$$

$$1 + (x')^2 = 1 + \frac{16y^8 + 1 - 8y^4}{16y^4} = \frac{16y^4 + 16y^8 + 1 - 8y^4}{16y^4}$$

$$= \frac{16y^8 + 8y^4 + 1}{16y^4} = \frac{(4y^4 + 1)^2}{16y^4}$$

$$L = \int_1^3 \sqrt{\frac{(4y^4+1)^2}{16y^4}} = \int_1^3 \left(\frac{4y^4 + 1}{4y^2}\right) dy$$

$$= \int_1^3 \left(y^2 + \frac{1}{4y^2}\right) dy = \int_1^3 \left(y^2 + \frac{y^{-2}}{4}\right) dy = \frac{y^3}{3} + \frac{y^{-1}}{(-1)4}\Bigg|_1^3$$

$$= \frac{y^3}{3}\Bigg|_1^3 - \frac{1}{4y}\Bigg|_1^3$$

$$= \left(\frac{3^3}{3} - \frac{1}{3}\right) - \left(\frac{1}{4(3)} - \frac{1}{4}\right) \quad \text{Ans}$$

$$= 9 - \frac{1}{12} - \frac{1}{3} + \frac{1}{4} = \boxed{\frac{53}{6}}$$